AUTOMATED VEHICLES

ENP-0112

3/11/2026

Dr. Divya C. Chandra

Yellowstone National Park Visitor Buses (Sumer 2021)

Figure 4: Exterior of Local Motors Olli 1.0 Shuttle

Source: U.S. DOT Volpe Center

PLAN

- Driving tasks and vocabulary
- Automation related to driving
- Flying vs. Driving comparison – class discussion

DRIVING

Tasks and Vocabulary

WHAT DOES IT MEAN TO “DRIVE”?

- DDT – Dynamic Driving Task
  - Lateral and longitudinal control
- OEDR – Object and Event Detection and Recognition
  - Active safety assistance, such as Automatic Emergency Braking Technology, AEBT
- ODD – Operating Design Domain
  - Getting into and out of the ODD

Most people think of themselves as **either the driver or a rider.**
What other role(s) are there?

AUTOMATED-SYSTEMS FOR DRIVING

Earlier Systems

- Automatic transmission
- Power steering
- Electronic vehicle stability (EVS)
- Antilock brakes (ABS)

Newer Driver Assistance Technologies (Level 1)

- Momentary Driver Assistance
  - Automatic emergency braking technology (AEBT)
  - Forward collision warning (FCW)
  - Lane departure warning
- (Continuous) Driver Assistance
  - Adaptive cruise control (ACC)
  - Lane keeping assistance
  - Blind spot monitoring
- Adaptable features (headlights)

CONFUSING TERMS (A FEW EXAMPLES)

- Similar names but different concepts
  - Remote assistance, remote driving, remote supervision, remote monitoring

    ("Driverless" does not mean there is no human in the system…)
- Different acronym but same idea
  - ADAS [Advanced Driver-Assistance System]**
  - ADS [Automated Driving System]**
- Marketing-speak
  - Fully self-driving, or even self-driving*

**Not to be confused with DADDS (Driver Alcohol Detection System for Safety)

*A 2018 automation survey from the MIT Age Lab found that 22.7% of respondents said that self-driving vehicles were available for purchase and 8.2% said yes to "Have you, or has someone you know, ever driven or taken a ride in a vehicle you would describe as self-driving?"

SAE LEVELS OF AUTOMATION

- SAE International
  - Standards for aviation and automotive companies
  - Industry-Recommended practices
  - https://www.sae.org
- How is the document (J3016) used?
  - Voluntary compliance
- Many factors are not addressed, e.g.,
  - Standardization of vehicle behavior across vehicles/manufacturers
  - Consumer education
  - Maintenance of vehicle safety features
  - Cost of the vehicle features
  - Cost of insurance
  - Risk of driver distraction with more automation?

SAE J3016™ LEVELS OF DRIVING AUTOMATION™

Learn more here: sae.org/standards/content/j3016_202104

Copyright © 2021 SAE International. The summary table may be freely copied and distributed AS-IS provided that SAE International is acknowledged as the source of the content.

	SAE LEVEL 0™	SAE LEVEL 1™	SAE LEVEL 2™	SAE LEVEL 3™	SAE LEVEL 4™	SAE LEVEL 5™
What does the human in the driver's seat have to do?	**You are driving** whenever these driver support features are engaged – even if your feet are off the pedals and you are not steering			**You are not driving** when these automated driving features are engaged – even if you are seated in "the driver's seat"		
	You must constantly supervise these support features; you must steer, brake or accelerate as needed to maintain safety			When the feature requests, you must drive	These automated driving features will not require you to take over driving	
	These are driver support features			**These are automated driving features**		
What do these features do?	These features are limited to providing warnings and momentary assistance	These features provide steering **OR** brake/ acceleration support to the driver	These features provide steering **AND** brake/ acceleration support to the driver	These features can drive the vehicle under limited conditions and will not operate unless all required conditions are met		This feature can drive the vehicle under all conditions
Example Features	• automatic emergency braking • blind spot warning • lane departure warning	• lane centering **OR** • adaptive cruise control	• lane centering **AND** • adaptive cruise control at the same time	• traffic jam chauffeur	• local driverless taxi • pedals/ steering wheel may or may not be installed	• same as level 4, but feature can drive everywhere in all conditions

Copyright © 2021 SAE International.

TRANSLATION

Driver handles OEDR
Driver is the fallback

0. No automation. Driver is totally involved.

1. Driver assistance – system handles steering *or* acceleration. Driver monitors.

2. Partial driving automation – system handles steering *and* acceleration. Driver monitors.

System handles OEDR

3. Conditional driving automation – human must monitor and respond. Driver is "fallback ready."

4. High driving automation – full automation in limited contexts (ODDs). Driver has no role when vehicle is in its ODD.

5. Full driving automation – full automation in *all* contexts. Driver has no role unless desired.

System is the fallback.

OVERSIGHT

- Approval of test sites is by State Department of Transportation (DOT)
- Federal Agencies involved
  - US DOT
    - https://www.transportation.gov/AV
  - National Highway Traffic Safety Administration (NHTSA)
    - https://www.nhtsa.gov/vehicle-safety/automated-vehicles-safety
  - Federal Transit Administration for public transit (e.g., buses, trains)
    - https://www.transit.dot.gov/research-innovation/strategic-transit-automation-research-plan-report-0116
  - Federal Motor Carrier Safety Administration (trucks)
    - https://www.fmcsa.dot.gov/research-and-analysis/technology/automated-cmv-evaluation-ace-program
- Some Federal laws that apply
  - Federal Motor Vehicle Safety Standards
  - American's with Disabilities Act (ADA)
  - Buy America Act

MITIGATIONS RELATED TO AUTOMATED VEHICLES

Individual

- Selection and training (Knowledge, Skills, and Abilities)
- Motivation, arousal (optimize attention)

Wide user population, relatively little training and oversight

Organization

- Policies and procedures (e.g., checklists)
- Safety culture

What is the "culture" of a driver? How can it be influenced?

System Design

- Controls (soft or hard), "Usability" (e.g., visibility of system state, interaction with data)
- Automated systems? (take the operator out of the system)

Operational Risk

- Variation across each operation (e.g., across flights)
- Work as imagined (theory) vs. work as done (reality)

AV designers can address these

AUTOMATION AND DRIVING

More on April 8 with Guest Lecturer Prof. Shannon Roberts (UMass)

WHY AUTOMATE?

- Reduce workload
  - Car cruise control, wing-leveler, heading hold
  - Offload “boring” tasks
  - *Make task easier, more comfortable*
- Improve safety
  - Perform tasks that are more ‘difficult’ or dangerous for humans
- Extend human capabilities
  - More efficiency, precision, flexibility
- Reduce costs
  - Reduce staff…~~e.g., flight engineer~~ or longer duty hours allowable for AVs?
- Because we can…?

From 1/26/26 lecture on automation in aviation

"USE" OF AUTOMATION

- ***Use:*** Voluntary activation or disengagement of automation by human operators
- ***Misuse:*** Overreliance on automation, which can result in failures of monitoring or decision biases
- ***Disuse:*** Neglect or underutilization of automation
- ***Abuse:*** Automation of functions without due regard for the consequences for human performance

Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. *Human Factors, 39*, 230-253.

CALIBRATED TRUST IN AUTOMATION

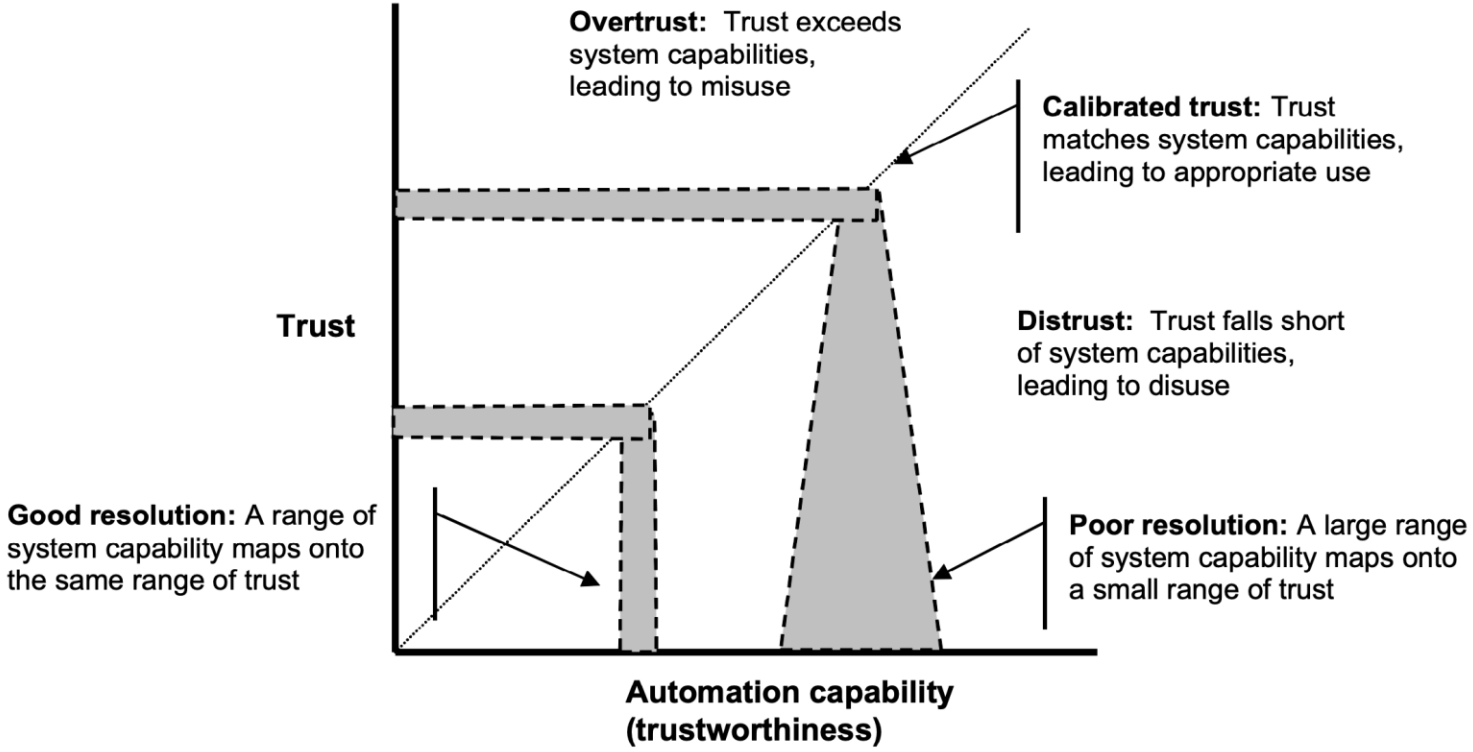

Figure 2. The relationship among calibration, resolution, and automation capability in defining appropriate trust in automation. Overtrust may lead to misuse and distrust may lead to disuse.

Lee J, See K. Trust in automation: Designing for appropriate reliance. *Human Factors* 46(1):50-80, 2004.

SOME POTENTIAL CONSEQUENCES OF ~~FLIGHT DECK~~ CAR AUTOMATION

- Boredom, complacency, lack of appropriate response in an emergency
- Misunderstanding or lack of knowledge of what automation is doing/considering
- Error in data entry/interaction
- Overconfidence in automation
- False alarms
- Mode confusion

(1/26/26 lecture)

Advantages & Disadvantages

16.400/453

Advantages	Disadvantages
• Increased capacity & productivity • Reduction of manual workload • Reduction of fatigue • Error prevention • Precision • Economy of design space	• Dehumanizing • Complacency • False alarms • Increase in mental workload • Automation-induced failures
Questionable	**Unknown**
• Overall workload reduction? • Operational cost reduced? • Training requirements reduced? • Reduction in crew size?	• Maintenance • Hardware & software costs • Redundancy • Liability issues • Personnel issues

How do these look today?

There are still legal and other unknowns.

What else?

(1/26/26 lecture)

EVOLVING AUTOMATION PHILOSOPHIES

- The automated system has final authority.
  - Keep the ~~pilot~~/driver from doing something foolish.
- The ~~pilot~~ driver has final authority.
  - Keep the automation from doing something foolish.
- Something in between?
  - Adaptive or adaptable automation?
  - Sharing authority?
  - Team work?

(1/26/26 lecture)

THOUGHTS ABOUT AUTOMATION

- Capabilities will continue to improve
- Use of automation/autonomy is intended to reduce cost and staffing and improve safety, but the opposite can occur
- Moving from demonstration to real world is challenging
- Creeping complexity
- Exhaustive testing of complex systems is not possible
- Bainbridge, Ironies of Automation, 1983

  "…the more advanced a control system is… the more crucial may be the contribution of the human operator."
- *Reimer, 2023 Webinar*
  - *"Robots make mistakes too."*
  - *"Electrification of vehicles provides performance characteristics and incredible acceleration."*
  - *"Automation and electrification offer opportunities for a better system, but they need to be combined together with human understanding and human expertise."*

DRIVING VS. FLYING

DIMENSIONS FOR COMPARING FLYING VS. DRIVING

Driving

Flying

MANAGING RISK: OPERATIONAL COMPLEXITY IN ~~FLIGHT OPERATIONS –~~ DRIVING: THE "ODD"

ATC Intervention (such as)
- (Late) route amendments
- Unpublished restrictions
- Vectors

Operator Factors
- Independence vs. dependence on Dispatch
- Clarity and consistency of pilot roles in reviewing route

Operational Complexity
- ATC interventions
- ~~Aircraft~~ Vehicle equipment or performance
- ~~Crew~~ Driver factors
- Operator factors
- Environment factors

~~Aircraft~~ Vehicle Factors
- Lack or unreliability of automated systems
- Performance characteristics

Environment Factors
- Terrain
- Traffic
- Weather
- Prohibited ~~airspace~~ roads

~~Crew~~ Driver Factors
- Expectations (confirmation bias)
- Fatigue
- Communication style
- Distractions
- Local area familiarity
- Familiarity with different types of routes

(2/18/26 lecture)

MORE DRIVING VS. FLYING

Driving

- Relatively little training or currency checks
- Wider population of operators
- Could stop and reset/review
- “Operational design domain” (ODD)
- Safety? (Private vs. Public risk)

Flying

- Intensive training and currency reviews
- Relatively limited set of operators
- Certifications and approval of both pilots and aircraft
- ATC monitors flights and can aid in emergencies
- Always in motion
- Public expects high safety levels for commercial operations

LINKS AND RESOURCES

- SAE
  - https://www.sae.org
  - https://www.sae.org/news/2021/06/sae-revises-levels-of-driving-automation
- Driving
  - The Simpsons - Robotic truck driving video
  - Tesla marketing:
    - Autopilot and Full Self-Driving Capability.
    - https://www.tesla.com/support/autopilot
- National Highway Traffic Safety Administration (NHTSA)
  - https://www.nhtsa.gov/technology-innovation/automated-vehicles-safety
  - https://www.nhtsa.gov/vehicle-manufacturers/automated-driving-systems
  - https://www.transportation.gov/briefing-room/trump-administration-releases-"ensuring-american-leadership-automated-vehicle
  - https://www.transportation.gov/AV
- MIT Age Lab, Transportation
  - https://agelab.mit.edu/transportation-and-livable-communities/overview/
  - Video from Sept 20, 2023 "ADAS in the Wild"
    - https://youtu.be/DUaYzxZsAig?feature=shared